AMRITSAR GROUP OF COLLEGES NAAC Grade "A" 3rd Cycle | Autonomous College under Autonomous Category | (Since 2014) Conferred by UGC	AGCS-21704 Major Project Lab	Department of Computer Science and Engineering

B.Tech (CSE) 7th Sem – 3 (Group G1)

FORTNIGHTLY PROJECT PROGRESS REPORT-4

Date : 01-09-2026 to 15-09-2026
Group number : 07

Project Title : Nexora – An Intelligent Knowledge Workspace

Project Members:

S.No	Univ. Roll No	Name	Role
1	2333428	Shivam Kumar	Member
2	2333429	Shivam Kumar Pandit	Member
3	2333436	Sonu Kamat	Syndicate Leader

Progress Summary	1. Developed the Sparse BM25 Keyword Search Engine for processing exact keyword matches, domain acronyms, and formulas. 2. Implemented Reciprocal Rank Fusion (RRF) algorithm to combine sparse BM25 retrieval with ChromaDB dense vector embeddings. 3. Configured fusion scoring with rank constant k=60 to balance exact keyword precision with dense semantic search. 4. Integrated Cross-Encoder dynamic chunk reranker to re-order top candidate chunks before passing context to LLM agents. 5. Implemented in-memory LRU query embedding cache to avoid redundant embedding calculations for frequent user queries. 6. Reduced overall RAG retrieval latency by 45%, achieving average response retrieval times under 180ms. 7. Conducted retrieval accuracy benchmarking using MRR@10 (Mean Reciprocal Rank) and NDCG@5 evaluation metrics. 8. Updated VectorStore module (backend/vector_store.py) to support hybrid search and score normalization. 9. Added dynamic score normalization to handle heterogeneous document chunk lengths across multi-page files. 10. Updated FastAPI backend REST API endpoints to support hybrid search query parameters and filtering options.
Work In Progress	1. Implementation of JWT-based user authentication and access control. 2. Designing multi-tenant private workspace isolation for document collections. 3. Development of browser-native Web Speech API voice querying interface. 4. Integration of NotebookLM-style 2-host audio podcast dialogue synthesizer. 5. Testing hybrid search precision across large document sets (100+ pages).

Challenges And Issues	1. Balancing weights between sparse BM25 scores and dense vector similarity scores for diverse document queries. 2. Managing computational overhead caused by Cross-Encoder reranking on large candidate chunk sets. 3. Normalizing BM25 scores across documents with varying chunk lengths and word frequencies. 4. Preventing cache invalidation issues when new documents are uploaded to the vector store. 5. Ensuring sub-second response times during hybrid retrieval without sacrificing search recall.